FOUR EQUIVALENT CONDITIONS FOR A MORPHISM FROM THE SPECTRUM OF A FIELD TO BE OF FINITE TYPE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- A morphism from the spectrum of a field is quasi-compact because its source is a singleton, so finite type and locally finite type agree for it. Used at proof lines 3059-3061. Gives the equivalence of the first two conditions.
- The algebra lemma that a finite type domain extension inside a field has a localization that is a field with finite extension of the target field, cited from the algebra chapter. Cited at proof lines 3068-3070. Produces the element whose localization turns the finite type injection into a finite field extension, giving the third condition.
- The same-chapter characterization of locally finite type morphisms on affine charts, and the algebra lemma that the image of a finite ring map into a field is a field over which the target is finite. Cited at proof lines 3074-3075 and 3079-3080. Give the implication from the third to the second condition, and the identification of the kernel of a finite ring map into a field as a maximal ideal.

Source reference. Original lemma 00FY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 588–588. Same-chapter reference used in the retained author text at source line(s) 2852. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00GR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 617–617. Same-chapter reference used in the retained author text at source line(s) 2869. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01T2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2359–2359. Same-chapter reference used in the retained author text at source line(s) 2717, 2739, 2801, 2863, 3019, 3035, 3157, 3348, 3613, 3707, 3868, 5197, 5282, 6702, 9325, 9390, 9440, 11421, 13422, 13686. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let S be a scheme. Let k be a field. Let $f : \mathrm{Spec}(k) \to S$ be a morphism. The following are equivalent:*

1. *The morphism f is of finite type.*
2. *The morphism f is locally of finite type.*
3. *There exists an affine open $U = \mathrm{Spec}(R)$ of S such that f corresponds to a finite ring map $R \to k$.*
4. *There exists an affine open $U = \mathrm{Spec}(R)$ of S such that the image of f consists of a closed point u in U and the field extension $k/\kappa(u)$ is finite.*

Proof. The equivalence of (1) and (2) is obvious as $\mathrm{Spec}(k)$ is a singleton and hence any morphism from it is quasi-compact.

Suppose f is locally of finite type. Choose any affine open $\mathrm{Spec}(R) = U \subset S$ such that the image of f is contained in U, and the ring map $R \to k$ is of finite type. Let $\mathfrak{p} \subset R$ be the kernel. Then $R/\mathfrak{p} \subset k$ is of finite type. By Algebra, Lemma 00FY there exist a $\overline{f} \in R/\mathfrak{p}$ such that $(R/\mathfrak{p})_{\overline{f}}$ is a field and $(R/\mathfrak{p})_{\overline{f}} \to k$ is a finite field extension. If $f \in R$ is a lift of $\overline{f}$, then we see that k is a finite R_f-module. Thus (2) $\Rightarrow$ (3).

Suppose that $\mathrm{Spec}(R) = U \subset S$ is an affine open such that f corresponds to a finite ring map $R \to k$. Then f is locally of finite type by Lemma 01T2. Thus (3) $\Rightarrow$ (2).

Suppose $R \to k$ is finite. The image of $R \to k$ is a field over which k is finite by Algebra, Lemma 00GR. Hence the kernel of $R \to k$ is a maximal ideal. Thus (3) $\Rightarrow$ (4).

The implication (4) $\Rightarrow$ (3) is immediate. $\square$